

Abhinav Saxena

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Sydney - Australia

PROFESSIONAL SUMMARY

I am a Computer Science graduate with a strong foundation in full-stack development, software engineering, and cloud-based systems. With hands-on experience in technologies such as React, Django, Java, Flutter, and AWS, I have developed and deployed scalable applications across web and mobile platforms. My strengths include designing efficient APIs, building responsive UIs, and working with tools and frameworks like Spring Boot, Firebase, Docker, and TensorFlow. I have also contributed to research in deep learning and served as Technical Head of the Ciphers Club, where I led several impactful initiatives.

My career goal is to pursue advanced studies and research in Computer Science, with a focus on intelligent systems and scalable software solutions. I aim to contribute meaningfully to technological innovation through a blend of academic rigor and real-world experience.

EDUCATION

- **The University of Sydney** Feb 2024 – April 2026
Masters in Computer Science (Advance Entry) Sydney, Australia
 - GPA: In Progress
- **Bennett University** Oct 2020 – June 2024
B.Tech in Computer Science and Engineering Greater Noida, India
 - GPA: 8.99/10.00
- **Delhi Public School, Kalyanpur** 2020
Senior Secondary Education (Class XII) – PCM Kanpur, India
 - Percentage: 82%
- **Delhi Public School, Kalyanpur** 2018
Secondary Education (Class X) Kanpur, India
 - CGPA: 9.2/10

EXPERIENCE

- **Esprit Analytique**  June 2024 – Present
Full Stack Developer & Developer Team Lead Remote
 - As a founding engineer, initiated a full-stack project (React + Python) and led it to completion as a production-ready website with a seamless and intuitive user experience.
 - Led a team of developers, overseeing project architecture, code reviews, and deployment workflows to ensure high-quality deliverables and alignment with business goals.
 - Designed and developed responsive web applications using React and Django with PostgreSQL, including multilingual support for several Indian languages.
 - Engineered and optimized 65+ RESTful APIs using Django and multithreading, achieving 30% performance improvement; integrated Razorpay for secure payment workflows.
 - Implemented RSA encryption for secure key exchange and AES for end-to-end payload encryption/decryption, ensuring robust frontend-backend data security.
 - Deployed scalable web applications using Docker on AWS EC2, ensuring smooth CI/CD and 99.9% uptime under peak traffic.
 - Built automation scripts for localization—translating JSON files, documents, and extracting strings from TSX files—streamlining the i18n process.
- **NEC Corporation India Pvt Ltd**  Oct. 2023 – Apr. 2024
Java Developer Intern Noida, India
 - Solely responsible for revamping the client-facing website from scratch, delivering a fully redesigned, responsive front-end aligned with modern UI/UX principles.
 - Developed RESTful APIs using Java Spring Boot with 85% code coverage, ensuring scalable, maintainable backend services.
 - Integrated backend APIs with a new responsive design, increasing user engagement by 25% and enhancing performance across devices.
 - Gained hands-on experience with Docker through independent mini-projects, enhancing understanding of containerization and deployment workflows.

- Acquired foundational knowledge of microservices architecture and service communication strategies during cross-functional discussions and technical sessions.

• Wormos

June 2023 – Aug. 2023

Flutter Developer Intern

Remote

- Built pixel-perfect mobile UI from Figma designs using Flutter, achieving 90% design accuracy and ensuring consistent cross-platform user experience.
- Integrated Firebase for real-time data storage and secure authentication, improving app performance and data reliability.
- Implemented OTP-based phone authentication using Firebase Authentication to ensure secure user verification.
- Integrated Google Sign-In and Sign-Up functionality to streamline user onboarding and enhance accessibility.

PROJECTS

• Localeora

August 2026 - Present

React, Django REST Framework, PostgreSQL, Google Cloud Translation API, Azure AI Translator

- Architected an **AST-based localization platform** that extracts user-facing strings directly from **React, TypeScript, TSX**, and **HTML** source code using a dedicated **Node.js extraction engine** (**Babel** parser and **traverse**, **parse5/cheerio**), decoupled from a **Django REST Framework** API that owns orchestration and persistence.
- Designed a two-engine pipeline separating read-only string extraction from source-rewriting transformation via **ts-morph**, with each edit re-validated against a content hash of the source file before being applied, preventing silent corruption of the codebase.
- Implemented a **dry-run-to-apply** workflow in which proposed edits are rendered as a reviewable diff and only written to disk after explicit approval, backed by structured, schema-validated artifacts shared across the Python and Node services.
- Built a provider-agnostic translation layer integrating **Google Cloud Translation** and **Azure AI Translator** behind a single interface, using **Python asyncio** with semaphore-bounded concurrency for batch translation jobs, polled asynchronously by the frontend for live progress.
- Developed the full-stack system end-to-end: a **Django REST Framework** and **PostgreSQL** backend with **JWT** and **OTP-based authentication**, and a **React, TypeScript**, and **Vite** frontend with **Tailwind CSS** and **React Three Fiber** for interactive 3D visualizations, supporting **30+ target languages**.

• GlassBox — Explainable Network Anomaly Detection System

Aug 2026 – Aug 2026

Tools: Python, Django REST Framework, PyTorch, scikit-learn, SHAP, scapy, React, TypeScript, Vite, SQLite

- **Decoupled ML pipeline** from serving: trained an Isolation Forest (**200 trees**, ~77 CICFlowMeter features) and a PyTorch **autoencoder** (input → 64 → 32 → 16-dim latent) on benign-only CIC-IDS2018 traffic, then serialized them as joblib/.pt artifacts loaded once at Django startup via a singleton ModelRegistry so per-request scoring never hits disk or re-parses models.
- **Honest evaluation and dual-model design**: thresholds tuned on a dedicated validation set while the held-out test set was touched exactly once by evaluate.py; autoencoder checkpointing uses **validation PR-AUC** (not benign reconstruction loss) to avoid the failure mode where the model gets better at reconstructing attacks too. Both scores are always returned so analysts can compare tree-based vs. neural signals per flow.
- **End-to-end SHAP explainability** wired into the real alert path: **TreeExplainer** for the Isolation Forest (exact, fast) and a **KernelExplainer** for the autoencoder, gated to fire *only* on flagged flows so the expensive model-agnostic path is not run on every clean packet; explanations are persisted as Explanation rows joined to each Alert so historical alerts stay re-analyzable if thresholds are later retuned.
- **Memory-safe preprocessing** of CIC-IDS2018 at scale: stratified train/val/test split, log1p on auto-detected non-negative features, **RobustScaler** fit on a 1M benign subsample, then **chunked scaling** in 100k-row blocks with ±10 clipping for the neural network — explicitly releasing intermediate frames via gc.collect() so the full pipeline fits on a laptop.
- **Live capture + hardened auth**: scapy AsyncSniffer assembles packets into flows and reuses the same analysis path as dataset replay; **X-API-Key** (HasAPIKey) enforced whenever DEBUG=False, a **10-second heartbeat file** (capture_status.json) distinguishes a real stopped capture from a crashed one, and threshold_multiplier exposes per-deployment recalibration as a first-class knob rather than a hack.

• Aarya's Spicy Food - Cloud Kitchen | [Visit](#)

May 2026

Tools: Next, React, TypeScript, Tailwind, Django REST, JWT, PostgreSQL, Firebase, Render, Supabase

- Developed a full-stack food ordering platform using **Next.js, React, TypeScript, Django REST Framework, JWT Authentication**, and **PostgreSQL**, delivering seamless customer and admin experiences.
- Built customer-facing features including **secure authentication, menu browsing, shopping basket management, order placement, real-time order tracking, order cancellation, and profile management**.
- Designed and implemented secure backend APIs with **JWT-based authentication, role-based access control (RBAC)**, customer-specific order history, order validation, and automated order status management.

- Developed a comprehensive **admin dashboard** enabling staff to efficiently **view, filter, update, and manage customer orders** through an intuitive real-time order management interface.
- Deployed the frontend on **Firebase Hosting**, backend services on **Render**, and managed the production database using **Supabase PostgreSQL**, ensuring a scalable and cloud-native deployment architecture.

• **Prodigidesk**

June 2024 - December 2025

Tools: React, Python, PostgreSQL, AWS EC2, Docker, Azure CI/CD, AES/RSA Encryption, JSON Localization

- Developed a scalable full-stack platform, **ProdiDesk**, using **React, Python**, and **PostgreSQL**, incorporating **JSON-based multilingual localization** to deliver seamless user experiences across multiple languages.
- Implemented a secure authentication and communication architecture using **JWT Authentication** and **Hybrid AES + RSA Encryption**, enabling secure API communication and supporting **100+ successful Razorpay payment transactions**.
- Built a custom **multi-process Python translation engine** for automated localization, improving translation throughput by **50%** while eliminating manual localization overhead in the deployment workflow.
- Containerized the application using **Docker** and deployed it on **AWS EC2**, improving scalability, portability, and production reliability.
- Leveraged **Google Cloud Platform (GCP)** for server-side rendering and metadata management, contributing to a **20% increase in SEO-driven organic traffic**.

• **Jamaican Patty House | Visit**

May 2026

Tools: React, TypeScript, Vite, Tailwind CSS

- Developed a responsive **single-page restaurant website** using **React, TypeScript**, and **Vite**, featuring a dynamic hero section, interactive menu catalog, business hours, and contact information.
- Implemented **scroll-based navigation**, animated reveal effects, and fully responsive layouts, ensuring a seamless user experience across mobile, tablet, and desktop devices.
- Designed structured **data models** for menu items and opening hours, enabling dynamic content rendering, simplified maintenance, and improved scalability.
- Crafted a custom **Jamaican-inspired UI/UX** with branded color palettes, typography, and interactive call-to-action components to deliver an engaging and visually consistent user experience.

• **Localis | Visit**

April 2025

Tools: React, TypeScript, Django, Vite, Tailwind, PostgreSQL, Axios, Google Translate API, Azure AI Translator, python-docx, asyncio

- Designed and developed a full-stack localization platform using **React (TypeScript)** and **Django REST Framework** to translate JSON language files and Word documents into **30+ languages**, including **22 Indian languages**, in a single batch operation.
- Integrated interchangeable **Google Translate** (deep-translator) and **Azure AI Translator** APIs, enabling flexible translation workflows based on cost, quality, and rate-limit requirements.
- Built an in-place **DOCX translation pipeline** using **python-docx** that rewrites document XML while preserving original formatting, styles, tables, and layout instead of converting content to plain text.
- Optimized large translation jobs using **Python asyncio** with semaphore-controlled concurrency (10 parallel requests) and implemented real-time job progress tracking through polling APIs, delivering faster processing and live frontend progress updates.
- Developed a responsive SPA with **React Router**, **Tailwind CSS**, and a custom glassmorphism UI, supporting multi-language selection, one-click ZIP downloads, and deployed the frontend and backend independently on Render with secure CORS communication.

• **PISCA**

November 2024

Tools: Java, Apache POI, Spring Boot, React, GIT

- Developed a responsive front-end interface for **PISCA** using **React**, improving user engagement by **20%** while reducing page load times by **15%**.
- Implemented **Excel report generation** using **Spring Boot** and **Apache POI**, enabling automated data export and streamlined reporting for stakeholders.
- Designed reusable and responsive UI components, ensuring a consistent user experience across desktop and mobile devices while improving application usability.
- Built backend reporting workflows with **Apache POI** to generate structured business reports, providing stakeholders with actionable insights for data-driven decision-making.

• **QR Code Generator**

October 2024

Tools: React, Tailwind, Typescript, Figma, Git

- Developed a responsive frontend for a **QR Code Generator** using **React**, enabling seamless PDF upload, processing, and QR code generation workflows.
- Implemented instant PDF preview functionality, providing users with real-time visual feedback before generating QR codes and improving overall usability.

- Built intuitive download and export features, ensuring a smooth end-to-end user experience for generated QR code assets.
- Optimized the frontend with reusable components and efficient state management, improving responsiveness, maintainability, and overall application performance.

- **Inbound Scanning**

March 2024

Tools: React, Springboot, Typescript, Tailwind, Git

- Developed separate **administrator** and **employee** dashboards using **React**, **TypeScript**, and **Tailwind CSS**, delivering role-specific interfaces with an intuitive user experience.
- Implemented secure **role-based authentication and authorization** using **Spring Boot**, ensuring protected access control and reliable admin- and employee-level permissions.
- Designed optimized workflows for administrators and employees, reducing user errors by **30%** through streamlined navigation and task-specific functionality.
- Built reusable UI components and responsive layouts, improving task efficiency and increasing overall task completion speed by **25%**.

SKILLS

- **Programming Languages:** Java, Python, C++, Dart, JavaScript, TypeScript
- **Web Technologies:** React, Next.js, HTML, CSS, Tailwind, Bootstrap, Node, Vite, Framer Motion
- **Database Systems:** PostgreSQL, MongoDB, MySQL
- **Data Science & Machine Learning:** TensorFlow, VGG-16, Deep Learning, Image Analysis
- **Cloud Technologies:** AWS (EC2), Azure, Google Cloud Platform (GCP), Firebase
- **DevOps & Version Control:** Docker, Git, GitHub, Gitea, Azure CI/CD, Firebase Hosting
- **Specialized Area:** AES Encryption, RSA, JWT, Hybrid AES + RSA Key Exchange, Proxy-based API Architecture
- **Backend Frameworks:** Spring Boot, Django, Express, Flutter
- **Other Tools & Technologies:** Postman, Razorpay, Apache POI, Google Translator API, Microsoft Azure Translator API, python-docx, asyncio, LaTeX
- **Research Skills:** Deep Learning Research, Technical Paper Writing, Problem-Solving, Cross-Functional Collaboration, Hackathon Development

HONORS AND AWARDS

- **4th Place**

Month Year

Smart India Hackathon (SIH)

- Secured 4th place among competing teams in a national-level hackathon focused on solving real-world problems
- Demonstrated strong problem-solving, rapid prototyping, and collaborative development skills under time constraints

- **Selected for Publication (Top 3)**

Month Year

Covid-19 Detection Using VGG-16

- Selected among a shortlisted group of 3 papers for publication, applying deep learning (VGG-16) techniques to Covid-19 detection via medical image analysis
- Recognized for applied machine learning expertise in AI-driven diagnostics research

LEADERSHIP EXPERIENCE

- **Lead Full Stack Developer**

June 2024 - Dec 2025

Esprit Analytique

- Promoted to Lead Full Stack Developer within one year after converting from a 3-month internship in just 17 days, based on outstanding performance
- Directed end-to-end architecture and development of ProdigyDesk across frontend, backend, and security layers
- Owned cloud deployment and DevOps strategy, leading containerization and CI/CD pipeline implementation for the team

- **Tech Head (Geek Cipher)**

Dec 2021 - Month Year

Ciphers Club, Bennett University

- Served as Tech Head of Ciphers Club, a core team leading the club's technical initiatives and student engagement
- Guided club members through technical projects and workshops, contributing to a collaborative learning environment
- Applied and strengthened technical leadership and mentorship skills while supporting the club's activities

VOLUNTEER EXPERIENCE

- **Event Host/Organizer**

Month Year

Bennett University — College Fest

- Hosted and organized a gaming tournament as part of Bennett University's annual college fest
- Managed event logistics, participant coordination, and rules/format for the competition
- Developed event management and public speaking skills while engaging a large student audience

CERTIFICATIONS

• HTML, CSS, and Javascript for Web Developers	July 2026
• Natural Language Processing with Classification and Vector Spaces	September 2023
• Machine Learning Engineering for Production (MLOps)	April 2023
• Graph Analytics for Big Data	April 2023
• Optimizing a Website for Google Search	March 2023
• Natural Language Processing with Sequence Models	September 2022
• Natural Language Processing with Probabilistic Models	September 2022
• Introduction to Artificial Intelligence (AI)	January 2022
• Machine Learning	November 2021
• Data Structures	November 2021

REFERENCES

1. **Mrs Sandhya Menon**

Co-Founder, Esprit Analytique

Email: sc3_menon@yahoo.co.in

Phone: References available upon request

Relationship: Direct Supervisor

2. **Mr Naveen**

Phone: References available upon request

Relationship: Mentor at NEC